Software Requirements Specification

for

VisionEdge: AI-Powered Optical Store Management & Virtual Try-On System

Version 1.0

Academic Context: Computer Science & Engineering - Major Project

Author: VisionEdge Development Team

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1. Introduction

1.1. Purpose

This document specifies the Software Requirements Specification (SRS) for VisionEdge, an end-to-end web platform designed to streamline optical store management, eye test recording, prescription digitization via Artificial Intelligence, face-shape frame recommendation, real-time interactive Virtual Try-On (VTO), inventory tracking, order fulfillment, and GST-compliant invoicing.

This SRS serves as a technical blueprint for project developers, academic advisors, evaluators, and system administrators. It rigorously details both functional and non-functional requirements in compliance with IEEE Std 830-1998 standards.

1.2. Scope

VisionEdge is a full-stack, AI-integrated web application comprising a Python Flask RESTful backend API and a responsive modern JavaScript/CSS frontend client. It provides a centralized digital ecosystem for optical retail management while introducing interactive computer vision tools directly to customers and staff.

The system will:

- Provide secure Role-Based Access Control (RBAC) supporting Admin, Staff/Optometrist, and Customer user roles via JWT token authentication.
- Maintain comprehensive Customer Relationship Management (CRM) including demographic records, medical history, eye condition profiles, and past prescriptions.
- Enable staff to record detailed clinical Eye Test parameters (OD/OS Sphere, Cylinder, Axis, Add Power, and Pupillary Distance).
- Digitally parse handwritten or printed prescription images using Google Gemini 1.5 Pro AI Vision OCR API, auto-extracting optical values into structured fields.
- Provide an interactive, canvas-based Virtual Try-On (VTO) engine allowing customers to preview spectacle frames live via webcam feed or uploaded photo.
- Execute automated AI recommendation models matching frame geometry to customer face shapes (Oval, Round, Square, Heart) and suggesting optical lens coatings (Blue-Cut, Anti-Reflective, Progressive, Photochromic) based on clinical prescription data.
- Manage dual inventory for spectacle frames and lenses, enforcing real-time stock reduction upon sales and low-stock alerts.
- Handle end-to-end order processing, payment status tracking, and automated PDF invoice generation with tax (GST) calculations.
- Host an AI Optical Assistant Chatbot for automated customer inquiry resolution.

The system will NOT (in version 1.0):

- Process direct credit card or online payment gateway transactions (transactions are registered as Cash, UPI, or Card at counter/POS).
- Require hardware-based automated refractometers or Bluetooth-connected optical devices (clinical values are entered manually or scanned via OCR).
- Perform clinical eye diagnostics or replace certified optometrists (AI outputs are advisory and subject to staff verification).

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
VTO	Virtual Try-On - an interactive computer vision feature rendering digital 2D/3D frames onto a facial overlay
OCR	Optical Character Recognition - technology used to extract text and numbers from optical prescription images
OD	*Oculus Dexter* - standard medical abbreviation for the Right Eye
OS	*Oculus Sinister* - standard medical abbreviation for the Left Eye
SPH	Sphere - optical lens power measured in diopters for nearsightedness or farsightedness
CYL	Cylinder - optical power for astigmatism correction
AXIS	Axis - orientation angle (0 to 180 degrees) of the cylinder power
ADD	Add Power - magnifying power added to the bottom of multifocal/progressive lenses
PD	Pupillary Distance - distance in millimeters between the centers of the pupils
RBAC	Role-Based Access Control - restriction of system access based on user roles (Admin, Staff, Customer)
JWT	JSON Web Token - secure compact URL-safe mechanism for transmitting claims between server and client
REST	Representational State Transfer - software architectural style for stateless web services
CRUD	Create, Read, Update, Delete - fundamental persistence operations on database entities

1.4. References

IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications*.

- Flask Documentation: https://flask.palletsprojects.com/
- SQLAlchemy & PyMySQL Documentation: https://docs.sqlalchemy.org/
- Google Gemini AI API Documentation: https://ai.google.dev/docs
- OpenCV / MediaPipe Vision Documentation: https://google.github.io/mediapipe/

"Digital Transformation in Optical Retail Management", International Journal of Computer Applications in Retail & Logistics*, 2024.

"Augmented Reality and Virtual Try-On Architectures for E-Commerce", IEEE Transactions on Human-Machine Systems*, 2025.

1.5. Overview

The remainder of this document is organized into seven key sections:

- Section 2 (Problem Statement) highlights current optical industry operational bottlenecks.
- Section 3 (Objectives) defines project goals.
- Section 4 (Literature Survey) reviews existing solution paradigms and research gaps.

- Section 5 (Requirements Specification) outlines detailed Functional, Non-Functional, User Role, and Interface requirements.
- Section 6 (Feasibility Analysis) evaluates technical, economic, legal, operational, and schedule viability.
- Section 7 (Technology Stack) details software libraries, frameworks, and deployment components.
- Section 8 (Data Dictionary) provides entity schemas and security access policies.

2. Problem Statement

Traditional optical retail stores and brick-and-mortar optometry practices suffer from significant operational inefficiencies and fragmented customer experiences. Conventional practices rely heavily on manual paper-based clinical notes, physical binder record-keeping, and disconnected point-of-sale cash registers.

Key challenges faced by current optical retailers include:

1. Manual Prescription Errors: Manual transcriptions of complex optical formulas (OD/OS sphere, cylinder, axis, prism) into sales orders lead to human error, incorrect lens fabrication, and customer dissatisfaction.

2. Lack of Automated Personalization: Customers often struggle to select appropriate spectacle frames that suit their physical face shape, leading to prolonged decision times and indecision.

3. Absence of Virtual Try-On Technologies: Brick-and-mortar stores limit try-ons to physical inventory in stock, while traditional e-commerce sites lack interactive preview tools to evaluate frame fit virtually.

4. Inventory & Prescription Fragmentation: Customer clinical eye exam records, lens stock levels, frame inventories, and billing invoices are managed across separate tools, creating stockouts, unfulfilled orders, and accounting overhead.

5. No Integrated Clinical-to-Sales Workflow: Optometrists lack a seamless tool to instantly push eye test measurements to the sales counter for frame recommendation, lens coating selection, and billing.

VisionEdge addresses this gap by synthesizing clinical eye test management, AI-driven OCR prescription extraction, face-shape recommendation algorithms, real-time webcam Virtual Try-On, inventory control, and automated billing into a single, cohesive software platform.

3. Objectives

- To design and implement a secure, scalable web-based optical store management system supporting multi-role authentication (Admin, Staff, Customer).
- To integrate an AI Vision OCR engine (Google Gemini 1.5 Pro) capable of extracting structured prescription parameters (OD/OS SPH, CYL, AXIS, ADD, PD) from image scans with high confidence scores.
- To build an interactive, browser-based Virtual Try-On (VTO) canvas application leveraging computer vision overlay techniques for real-time frame preview.
- To construct intelligent recommendation engines that evaluate face shapes (Oval, Round, Square, Heart) and prescription severity to suggest optimal frame geometry and lens coatings (Blue-Cut, Anti-Reflective, High-Index).
- To establish a unified database schema connecting customers, clinical appointments, eye tests, prescriptions, frame/lens stock, purchase orders, and PDF invoices.
- To enforce robust Role-Based Access Control (RBAC) ensuring data isolation and operational compliance across staff and administrative workflows.
- To evaluate system performance, UI responsiveness, and AI extraction accuracy through empirical validation in a simulated optical store environment.

4. Literature Survey

#	Study / Platform	Focus	Gap Identified
1	Generic Retail POS Apps (QuickBooks, Tally)	Standard retail inventory and billing management	Lacks specialized optical data structures (OD/OS prescription parameters, lens refractive index, frame dimensions, cylinder axis tracking).
2	Optical Store Software (MyVisionExpress, FlexOptical)	Legacy desktop-based optical clinical management	Heavy desktop footprint, expensive licensing fees, lacks web-native Virtual Try-On and AI prescription image extraction.
3	E-Commerce Eyewear Platforms (Lenskart, Warby Parker)	Consumer-facing online eyewear web/mobile applications	Proprietary closed ecosystems tailored solely to single-brand sales; non-adaptable as an in-store management platform for independent optical retailers.
4	"Augmented Reality Virtual Try-On Systems" (*IEEE*, 2024)	Computer vision facial landmark positioning for glasses fitting	Focuses purely on 3D rendering algorithms without connecting to clinical eye test data, inventory levels, or POS billing workflows.
5	"AI OCR in Medical Prescription Analysis" (*IJCA*, 2025)	Deep learning extraction of handwritten prescriptions	Addresses OCR accuracy in isolation without integrating into an active retail order pipeline or customer CRM.

Conclusion: Existing solutions are either narrow clinical desktop tools, generic retail POS systems lacking optical domain capabilities, or standalone AR research prototypes. VisionEdge fills this gap by uniting AI clinical vision tools, Virtual Try-On, retail inventory, and billing into an integrated web platform.

5. Requirements Specification

5.1. Introduction

- Purpose: A full-stack web application providing automated optical store administration, AI prescription scanning, facial frame recommendation, Virtual Try-On, and inventory billing.
- Scope: Clinical appointment booking, eye exam recording, Gemini AI OCR scanning, VTO canvas engine, frame/lens stock management, order processing, PDF invoicing, and role-restricted admin controls.
- Intended Users: Store Administrators, Optometrists, Sales Staff, and Retail Customers.

5.2. Functional Requirements

ID	Requirement	Description
FR1	User Authentication & RBAC	User registration and login utilizing JWT tokens, supporting Admin, Staff, and Customer user roles.
FR2	Customer CRM & Profile Management	Create, view, update, and search customer records, including demographic data, eye conditions, and medical history.
FR3	Appointment Scheduling	Schedule, manage, and filter customer appointments for Eye Tests, Frame Trials, Lens Consultations, and Follow-Ups.
FR4	Eye Test & Exam Recording	Record detailed clinical eye measurements (OD/OS SPH, CYL, AXIS, ADD, PD, doctor remarks) linked to customer records.
FR5	AI Prescription OCR Scan	Upload optical prescription images (JPG/PNG) and automatically extract numerical parameters using Gemini 1.5 Pro Vision AI.
FR6	AI Frame Recommendation	Detect or input face shape geometry (Oval, Round, Square, Heart) and return algorithmic frame recommendations.

ID	Requirement	Description
FR7	AI Lens Recommendation	Evaluate prescription power parameters and suggest suitable lens types (Single Vision, Progressive) and coatings (Blue Cut, Anti-Glare).
FR8	Interactive Virtual Try-On (VTO)	Render selected frame overlays live onto user webcam feed or static snapshot image via interactive canvas rendering.
FR9	Frame & Lens Inventory Control	Perform full CRUD operations on spectacle frames and optical lenses, tracking brand, model, material, price, stock, and min_stock levels.
FR10	Order Processing & Fulfillment	Generate sales orders linking customer, prescription, selected frame, and lens items; update fulfillment states (Pending, Processing, Ready, Delivered).
FR11	Billing & PDF Invoice Generation	Calculate subtotal, discounts, GST tax amounts, generate official PDF receipts, and update stock quantities automatically.
FR12	Analytics & Reports	Display administrative dashboards detailing total sales revenue, customer visit trends, low-stock warnings, and order status breakdowns.

5.3. Non-Functional Requirements

ID	Requirement	Standard / Target
NFR1	Response Performance	API endpoints respond within < 300ms under normal load; AI OCR extraction returns within < 3.5 seconds.
NFR2	Data Isolation & Security	Passwords hashed using Werkzeug bcrypt/PBKDF2; API routes protected by role-based JWT bearer authentication.
NFR3	Real-Time VTO Frame Rate	Virtual Try-On canvas maintains >= 24 FPS during live webcam video processing on standard client web browsers.
NFR4	Cross-Browser Compatibility	Web interface fully compatible with modern versions of Chrome, Edge, Firefox, and Safari across desktop and tablet viewports.
NFR5	Reliability & Data Integrity	Database operations execute within ACID compliant SQLAlchemy transactions; automatic error logging handles failovers.
NFR6	Scalability	Backend micro-architecture deployable via WSGI containers (Gunicorn / Waitress) behind Nginx reverse proxies.

5.4. User Roles

- Admin: Complete system authority. Manages staff credentials, oversees full inventory, accesses financial analytics, configures system settings, and overrides audit permissions.
- Staff / Optometrist: Handles front-desk and clinical operations. Conducts eye exams, scans prescriptions via AI OCR, manages appointments, assists with VTO, creates sales orders, and generates customer bills.
- Customer: Registered end-user. Views personal appointment history, browses frame catalogs, uses the Virtual Try-On tool, accesses saved prescription records, and checks order fulfillment statuses.

5.5. Assumptions and Constraints

- Assumptions:
- Users possess a web browser with camera access enabled for Virtual Try-On capabilities.
- An active internet connection is available to communicate with the Google Gemini AI API endpoints.
- In-store point-of-sale payments are settled at the counter (Cash, Card, UPI) before bill completion.
- Constraints:
- The initial release targets web browsers (desktop/tablet/mobile web); native mobile app wrappers are reserved for future phases.
- AI prescription parsing requires legible prescription images; low-resolution images trigger manual edit prompts for staff verification.

5.6. External Interface Requirements

5.6.1. User Interface

- Login & Auth Interface: Responsive form supporting secure login, role selection, and password recovery.
- Staff Dashboard & CRM: Tabular interface with instant search and filter controls for customer profiles and appointments.
- Clinical Eye Test & AI Scanner Module: Interactive form for OD/OS clinical data entry, with side-by-side file drag-and-drop preview for AI prescription image parsing.
- Virtual Try-On Suite: Dynamic web canvas integrating live camera controls, frame selector carousel, snapshot capture, and history comparison.
- Inventory & Order Management View: Grid layout showcasing frame stock, lens types, pricing metrics, stock alert indicators, and status badges.
- Billing & PDF Viewer Module: Order itemization table with automatic GST calculation and single-click PDF invoice downloading.

5.6.2. Hardware Interface

- Standard HD Webcam / Device Camera: 720p or higher video input for real-time Virtual Try-On facial tracking.
- Standard Display Monitor: Minimum resolution of 1024x768 pixels for desktop administration.

5.6.3. Software Interface

Interface	Description
Flask RESTful API	Serves JSON endpoints for all frontend client operations over HTTP/HTTPS.
SQLAlchemy / PyMySQL	Relational Database Abstraction Layer connecting Flask to MySQL or SQLite databases.
Google Gemini 1.5 Pro API	Multimodal AI Vision service used for optical prescription OCR parameter extraction.
Canvas / MediaPipe API	Client-side JavaScript library driving real-time webcam rendering for Virtual Try-On.
ReportLab / FPDF	Python server-side library generating formatted PDF billing receipts and invoices.

5.6.4. Communication Interface

- All client-server communication occurs securely via HTTP/HTTPS standard RESTful requests using JSON payloads.
- Media files (prescription uploads, VTO snapshots) are transmitted via multipart/form-data and served securely under token-validated /media/ static routes.

6. Feasibility Analysis

6.1. Technical Feasibility

The technologies selected-Python Flask, SQLAlchemy, MySQL, JavaScript Canvas API, and Google Gemini AI-are mature, industry-standard, and well-documented. The development team has demonstrated proficiency in building RESTful architectures and integrating computer vision algorithms. The system requires no specialized hardware beyond standard retail PC configurations and webcams, confirming complete technical feasibility.

6.2. Economic Feasibility

VisionEdge is economically advantageous. Built on open-source frameworks (Flask, SQLAlchemy, MySQL), the system

incurs zero initial software licensing overhead. Operational cloud deployment costs on PaaS platforms (e.g., Render, AWS EC2, digital VPS) combined with usage-based Gemini API pricing remain extremely modest ($5-$20/month for medium volume retail stores), yielding a highly favorable return on investment (ROI).

6.3. Legal Feasibility

VisionEdge complies with standard data privacy regulations. Customer medical notes and contact details are stored securely within protected databases using role-gated access. No biometric facial models are permanently sold or shared with third parties; VTO snapshots are stored locally or in secure app storage strictly for customer record history. No healthcare device certification is required as the platform functions as administrative management and decision-support software.

6.4. Operational Feasibility

The platform replaces fragmented paper-based processes with an intuitive web workflow. Front-desk staff and optometrists require minimal training due to clear user interface patterns mirroring traditional optical consultation steps (Appointment $\rightarrow$ Eye Exam $\rightarrow$ Frame VTO $\rightarrow$ Order $\rightarrow$ Invoice). High operational feasibility is assured.

6.5. Schedule Feasibility

The core functional system-including backend database modeling, RESTful API architecture, authentication, AI OCR scanning, Virtual Try-On canvas, inventory CRUD, and billing-has been modularly engineered and tested within a standard academic project timeline.

7. Technology Stack

Layer	Technology	Justification
Frontend UI	HTML5, Vanilla CSS3, JavaScript (ES6+)	Lightweight, zero-dependency, ultra-fast rendering with native browser Canvas API support.
Backend Framework	Python 3.9+ / Flask 3.0	Modular WSGI micro-framework offering lightweight RESTful routing and rapid extension integration.
Database ORM	SQLAlchemy 2.0 / Flask-SQLAlchemy	Robust relational mapping, transactional integrity, and automated model migrations.
Database Engine	MySQL 8.0 / SQLite 3	Enterprise-grade relational storage supporting ACID compliance, indexing, and foreign key constraints.
Authentication	Flask-JWT-Extended / PyJWT	Secure, stateless JSON Web Token bearer authentication for API role restriction.
AI Vision OCR	Google Gemini 1.5 Pro Vision API	Multimodal AI model providing high-precision structural extraction from prescription images.
Virtual Try-On Engine	HTML5 Canvas / MediaPipe Vision	Real-time browser-based computer vision rendering for frame overlay without server latency.
PDF Generation	FPDF / ReportLab	Programmatic creation of branded, printable customer GST billing invoices.
WSGI Server	Gunicorn / Waitress	High-concurrency production server suited for Linux VPS and Windows server deployments.
Version Control	Git & GitHub	Source code control, collaborative branching, and continuous delivery tracking.

8. Data Dictionary

The VisionEdge database is implemented in MySQL / SQLite using SQLAlchemy ORM models. All primary keys use auto-incrementing integers or UUID strings, enforcing strict relational integrity.

8.1. Table: user

Manages login credentials, staff account details, and role assignments.

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Unique system user ID
username	String(80)	Unique, Not Null	Account login handle
first_name	String(50)	Nullable	User's given first name
last_name	String(50)	Nullable	User's surname
email	String(120)	Unique, Nullable	Account contact email
password_hash	String(255)	Not Null	Bcrypt/PBKDF2 secure hashed password
phone	String(20)	Nullable	Contact telephone number
role	String(50)	Default 'customer'	User authorization role (admin, staff, customer)
created_at	DateTime	Default utcnow()	Account creation timestamp

8.2. Table: customer

Stores comprehensive retail customer profiles and demographic metadata.

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Unique customer ID
customer_code	String(20)	Unique	System generated customer reference code
first_name	String(50)	Not Null	Customer's first name
last_name	String(50)	Not Null	Customer's last name
gender	String(20)	Nullable	Customer gender identification
date_of_birth	Date	Nullable	Customer birthdate
phone	String(20)	Not Null, Indexed	Primary mobile phone number
email	String(120)	Indexed	Contact email address
address	Text	Nullable	Residential street address
city	String(100)	Nullable	Resident city
status	String(20)	Default 'Active'	Customer account status (Active, Inactive)
created_at	DateTime	Default utcnow()	Registration timestamp

8.3. Table: eye_test & prescription

Table: `eye_test`

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Unique eye exam record ID
customer_id	Integer	Foreign Key -> customer.id	Associated customer reference
appointment_id	Integer	Foreign Key -> appointment.id	Associated appointment (optional)
right_sphere	String(20)	Nullable	OD Sphere power value
right_cylinder	String(20)	Nullable	OD Cylinder power value
right_axis	String(20)	Nullable	OD Axis degree value
left_sphere	String(20)	Nullable	OS Sphere power value
left_cylinder	String(20)	Nullable	OS Cylinder power value
left_axis	String(20)	Nullable	OS Axis degree value
pd_value	String(20)	Nullable	Pupillary Distance (mm)
remarks	Text	Nullable	Optometrist clinical observations
created_at	DateTime	Default utcnow()	Examination timestamp

Table: `prescription`

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Unique prescription record ID
prescription_number	String(50)	Unique	Tracking reference number
customer_id	Integer	Foreign Key -> customer.id	Linked customer reference
eye_test_id	Integer	Foreign Key -> eye_test.id	Associated clinical eye exam record
add_power	String(20)	Nullable	Near Add power for progressive lenses
prescription_image	String(255)	Nullable	File path to uploaded scan image
ai_extracted_text	Text	Nullable	Raw JSON text parsed by Gemini AI
confidence_score	Float	Nullable	AI parsing confidence rating (0.0 to 1.0)
doctor_name	String(100)	Nullable	Prescribing eye doctor/optometrist name
created_at	DateTime	Default utcnow()	Record entry timestamp

8.4. Table: frame & lens

Table: `frame`

Column	Type	Constraints	Description

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Unique frame item ID
frame_code	String(50)	Unique	Frame SKU / Inventory code
brand	String(100)	Not Null	Frame manufacturer brand
model	String(100)	Not Null	Model designation
shape	String(50)	Not Null	Frame geometry (Oval, Rectangle, Round, Square, Cat-Eye)
color	String(50)	Not Null	Frame color description
material	String(50)	Nullable	Construction material (Acetate, Titanium, Metal)
price	Float	Not Null	Retail selling price (INR)
stock	Integer	Default 0	Current quantity available in store
min_stock	Integer	Default 5	Threshold for triggering low-stock alert
vto_image	String(255)	Nullable	Transparent PNG image URL for Virtual Try-On canvas

Table: `lens`

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Unique lens item ID
lens_code	String(50)	Unique	Lens SKU / Inventory code
brand	String(100)	Not Null	Lens manufacturer
type	String(50)	Not Null	Lens optical design (Single Vision, Bifocal, Progressive)
coating	String(100)	Nullable	Specialty coating (Anti-Glare, Blue Cut, Photochromic)
price	Float	Not Null	Retail selling price (INR)
stock	Integer	Default 0	Available lens blank stock quantity
min_stock	Integer	Default 10	Minimum stock threshold

8.5. Table: order & billing

Table: `order`

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Internal order system ID
order_id	String(50)	Unique	Customer-facing order code (e.g. ORD-1002)
customer_id	Integer	Foreign Key -> customer.id	Purchasing customer
prescription_id	Integer	Foreign Key -> prescription.id	Prescription applied to order
status	String(50)	Default 'Pending'	Order fulfillment stage (Pending, Processing, Ready, Delivered)
total_amount	Float	Default 0.0	Calculated order gross total
created_at	DateTime	Default utcnow()	Order placement timestamp

Table: `billing`

Column	Type	Constraints	Description
id	Integer	Primary Key, Auto Increment	Invoice record ID
invoice_number	String(50)	Unique, Not Null	Official legal invoice number (e.g. INV-2026-008)
order_id	Integer	Foreign Key -> order.id	Associated sales order
subtotal	Float	Default 0.0	Net item amount before taxes
discount	Float	Default 0.0	Total promotional discount applied
gst	Float	Default 0.0	Calculated Goods & Services Tax amount
total_amount	Float	Not Null	Final payable invoice balance
payment_status	String(20)	Default 'Pending'	Payment completion state (Paid, Pending)
payment_method	String(50)	Nullable	Payment tender mode (Cash, Card, UPI)
invoice_pdf	String(255)	Nullable	File path to generated PDF invoice

8.6. Role-Based Access Control (RBAC) & Security Policy Summary

Endpoint / Target	Required Role	Access Rule / Policy Description
/api/auth/register, /api/auth/login	Public	Open access for user authentication and token creation.
/api/customers/*	admin, staff	Staff and Admin users may search, view, create, and update customer profiles.
/api/ai/extract	admin, staff	Staff and Admin users can upload prescription images to invoke Gemini OCR.
/api/vto/frames, /api/vto/save-history	admin, staff, customer	Authenticated users can fetch VTO-ready frames and persist try-on snapshots.
/api/inventory/* (POST, PUT, DELETE)	admin	Only Admin users may add new frame/lens inventory items or modify prices/stock.
/api/billing/*	admin, staff	Staff and Admin users may issue bills, process payments, and download invoices.
/api/reports/*	admin	Strictly restricted to Admin accounts for viewing revenue analytics and audit logs.

End of Software Requirements Specification (SRS) Document for VisionEdge.